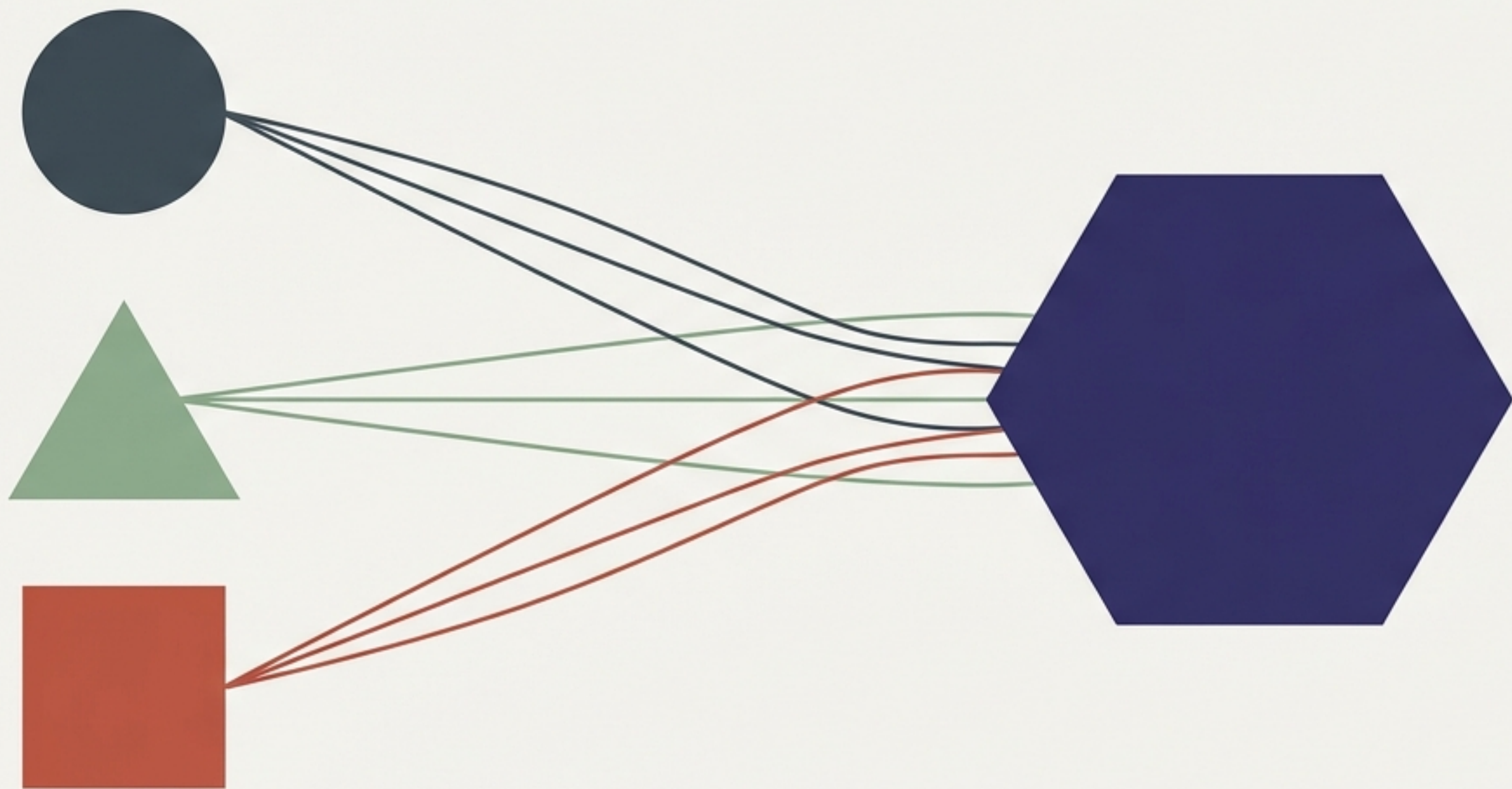


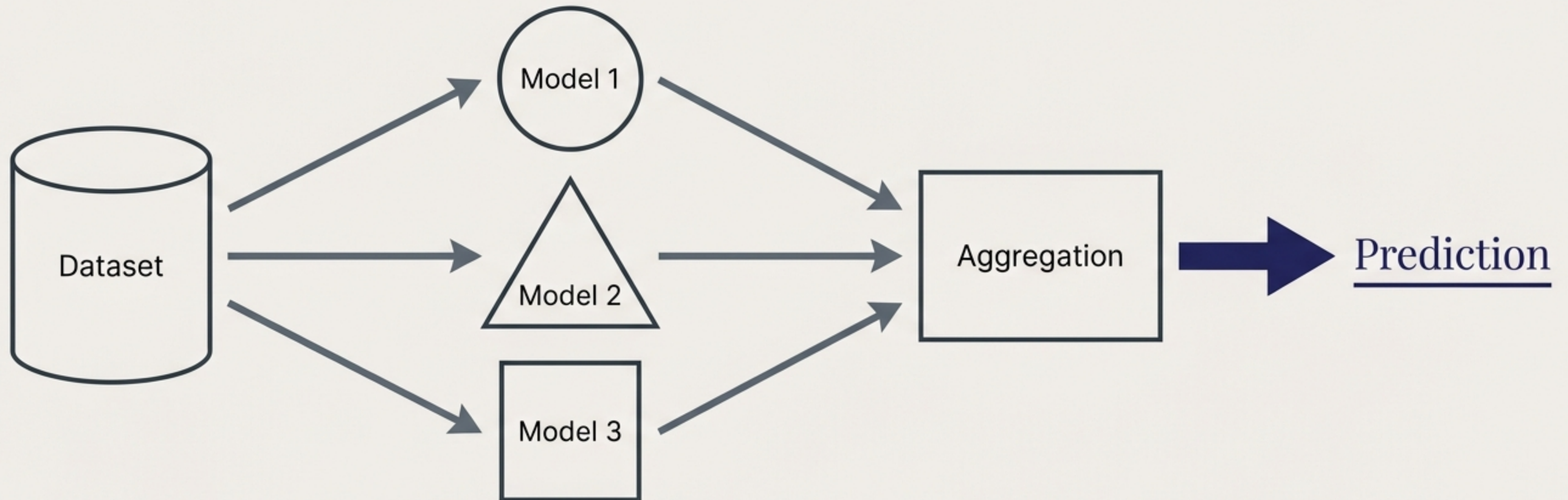
The Mathematics of Crowd Intelligence

Demystifying the Voting Ensemble algorithm and the probability that makes it work



The simplest ensemble technique relies on collective decision making.

A Voting Ensemble combines the predictions of multiple, independent base models to create a single, stronger meta-model. Rather than trusting one algorithm, it trusts the crowd.



The aggregation method depends on the task.

Classification

M1 votes 1

M2 votes 0

M3 votes 1



Majority Rules

-> Output: 1

Regression

M1 predicts 0.75

M2 predicts 0.92

M3 predicts 0.85



Calculate Mean

-> Output: 0.84

The Mathematical Paradox

If three separate models max out at an accuracy of 70%, how can combining them create a single model with an accuracy of 78.4%?

$$\begin{aligned} & [M_1 (70\%)] + \\ & + [M_2 (70\%)] + \\ & + [M_3 (70\%)] = 78.4\% ? \end{aligned}$$

The intuition lies in the Game Show Audience Poll.

Think of a TV game show audience. If the entire audience consists solely of software engineers, they will perfectly answer coding questions but collectively fail history questions. However, a highly diverse audience has scattered knowledge. When they vote, individual blind spots are masked by the varied expertise of the crowd.



Proving the paradox with probability.

Let's assume we have three independent models predicting a binary outcome. For the ensemble to win, the majority (at least 2 out of 3 models) must be correct.

Parameters

Base Accuracy: 70%

Probability of Correct (P_c) = 0.7

Probability of Incorrect (P_i) = 0.3

Win Condition: ≥ 2 models
predict correctly

Mapping the four paths to a correct majority vote.

Path	Model 1	Model 2	Model 3
Path 1	✓	✓	✓
Path 2	✓	✓	✗
Path 3	✓	✗	✓
Path 4	✗	✓	✓

Calculating the combined strength.

All 3 Correct

$$0.7 \times 0.7 \times 0.7 = 0.343$$

2 Correct, 1 Wrong (Happens 3 times)

$$0.7 \times 0.7 \times 0.3 = 0.147$$

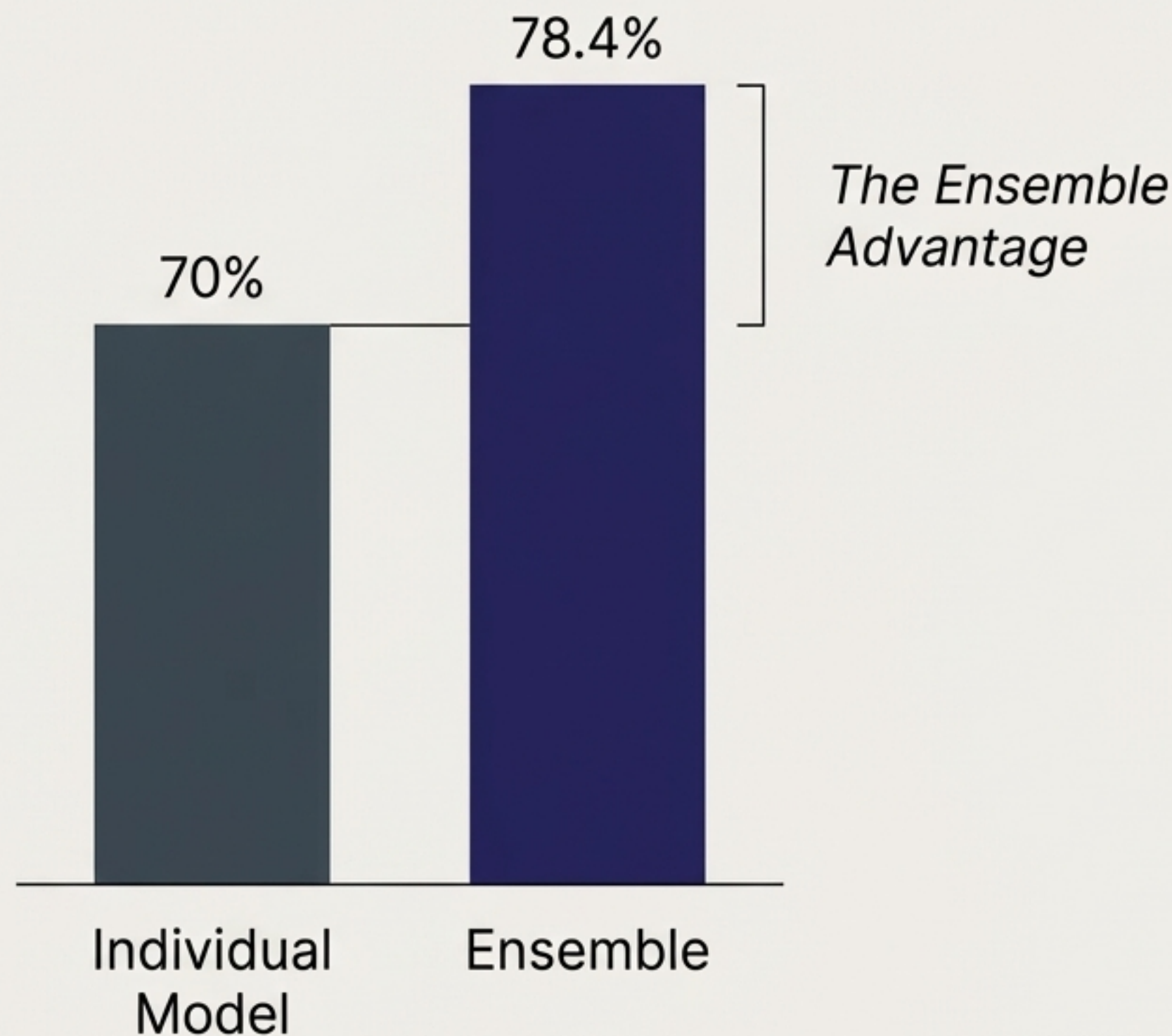
$$3 \times 0.147 = 0.441$$

Total Ensemble Accuracy

$$0.343 + 0.441 = \mathbf{0.784}$$

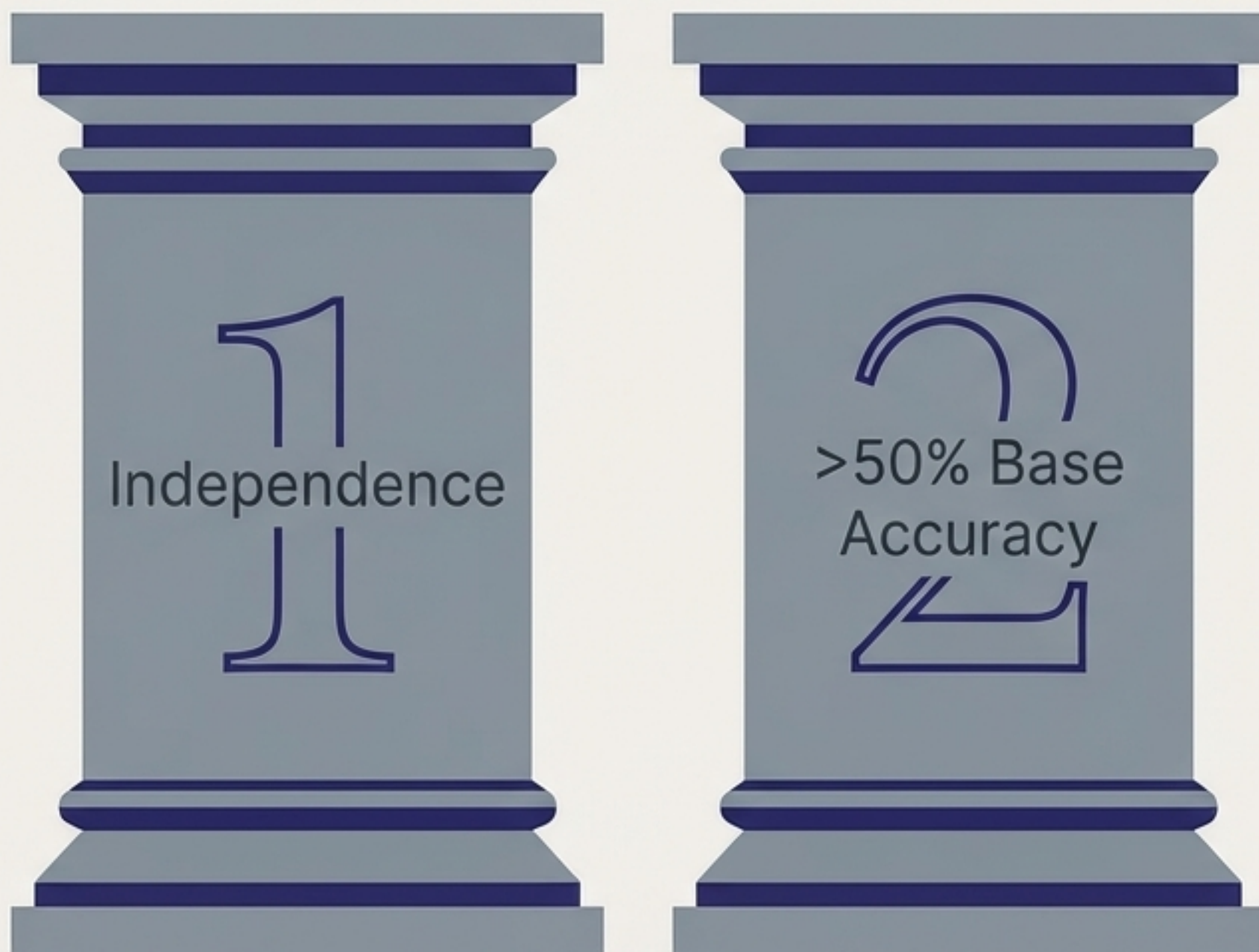
$$78.4\% > 70\%$$

The mathematics confirm it. The Voting Ensemble is objectively stronger than any of its individual components. Because the models are independent, their errors do not overlap perfectly, allowing the correct majority to constantly outvote individual mistakes.



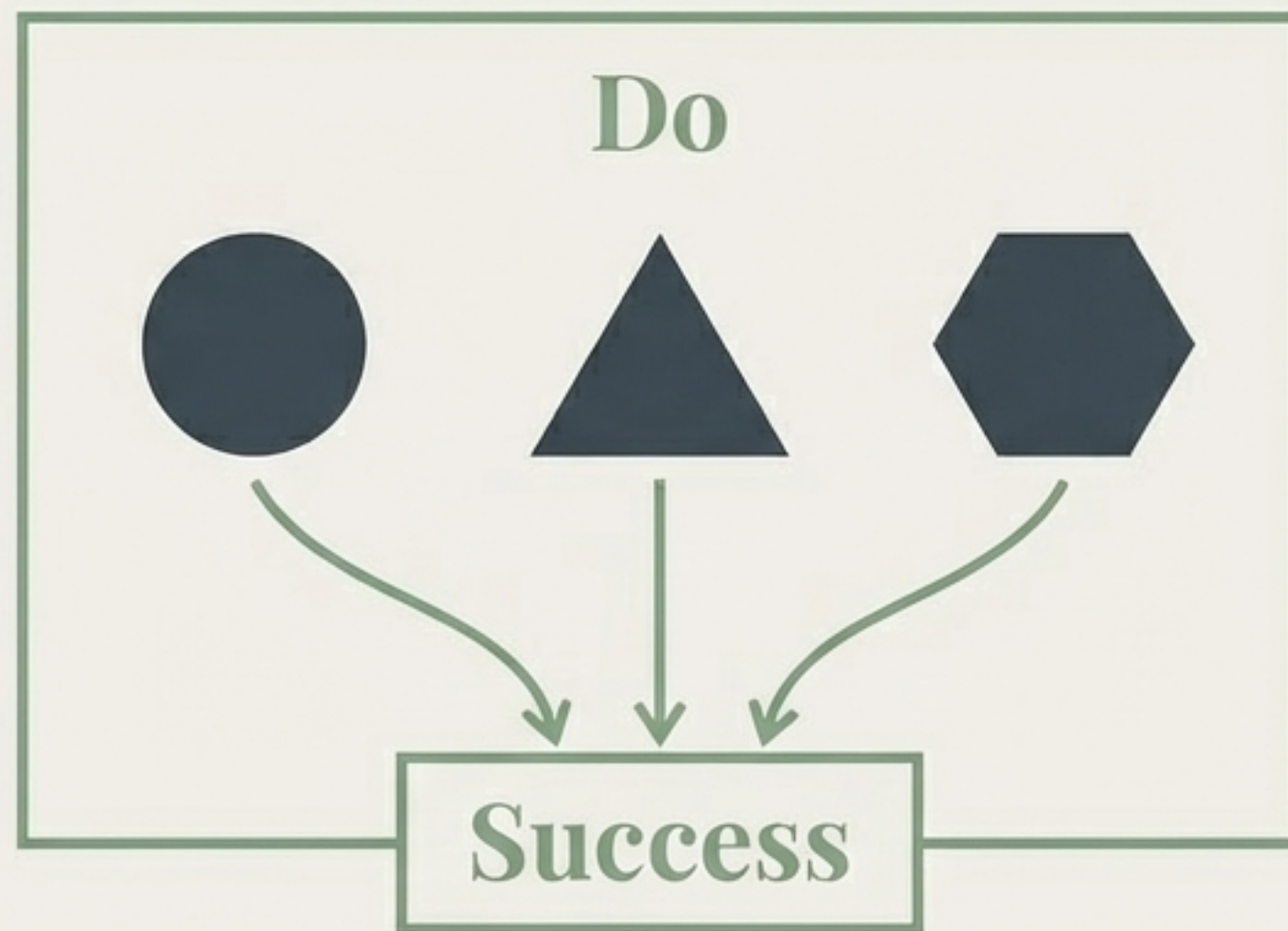
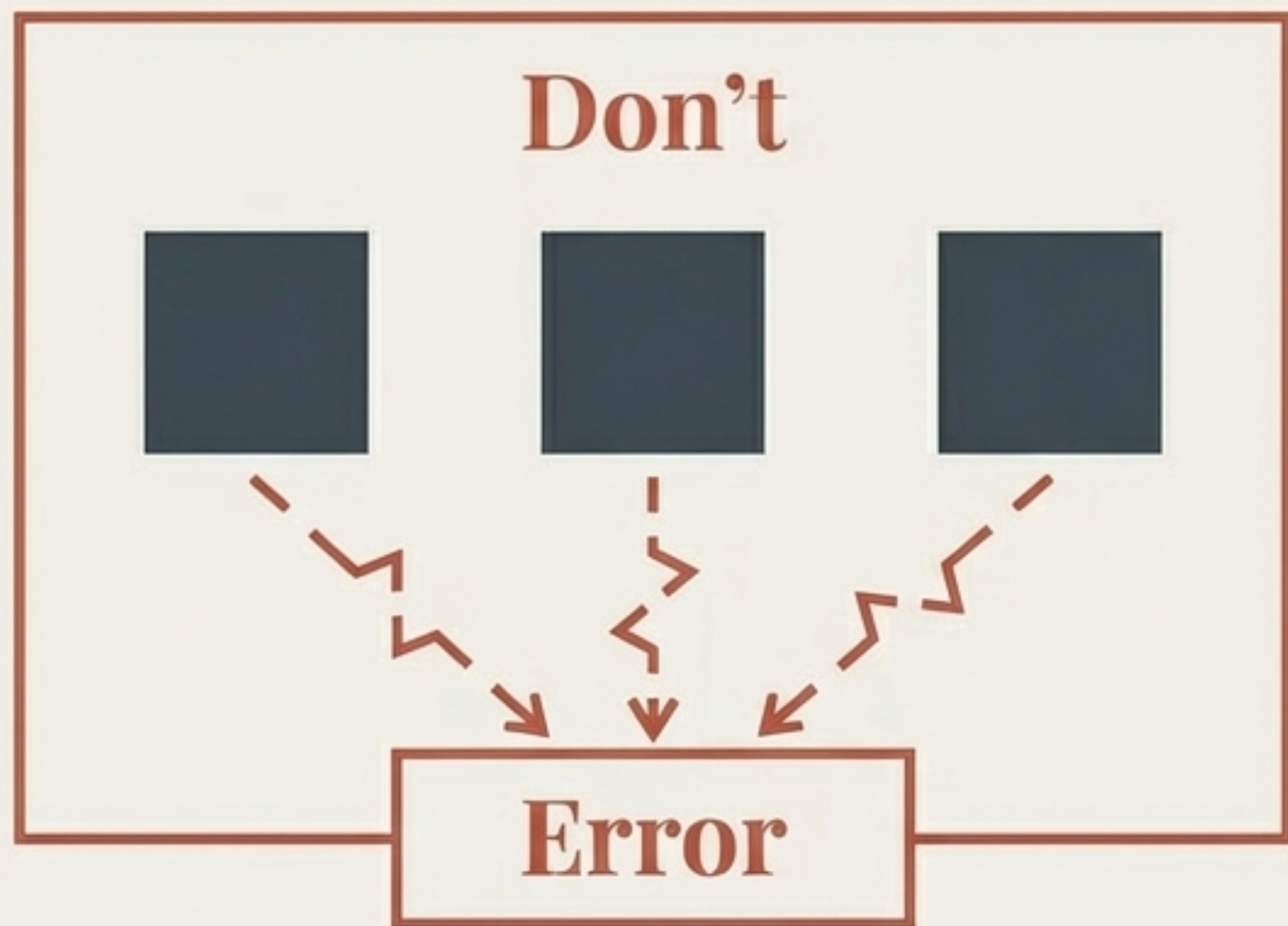
The magic only works under two strict rules.

A Voting Ensemble is not a guaranteed fix for bad data or poor modeling. To achieve this mathematical advantage, your base models must strictly adhere to two fundamental conditions.



Rule 1: The models must be independent.

If your models are identical or highly correlated, they will simply make the exact same mistakes at the exact same time. The ensemble will fail to capture the wisdom of the crowd and will just amplify the same errors.



Rule 2: Base accuracy must exceed 50%.

This is the critical twist. Individual models must be better than a random coin flip. If your base models are weak, the Voting Ensemble doesn't just fail to improve them—it actively destroys them. Let's look at what happens if base models have a 30% accuracy.



Voting amplifies failure when models are weak

Probability Correct = 0.3 | Probability Incorrect = 0.7

All 3 Correct

$$0.3 \times 0.3 \times 0.3 = 0.027$$

2 Correct, 1 Wrong (x3)

$$3 \times (0.3 \times 0.3 \times 0.7) = 0.189$$

Total Accuracy

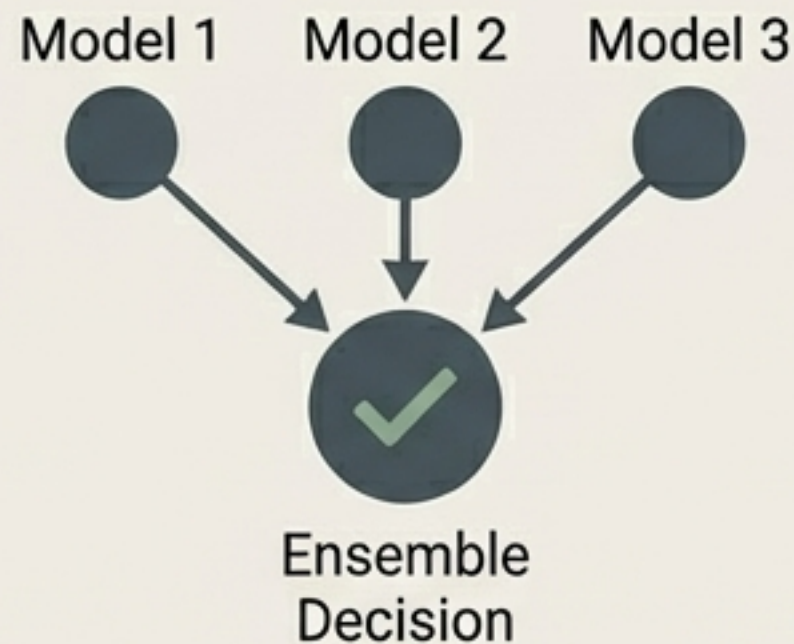
$$0.027 + 0.189 = 0.216 \text{ (21.6\%)}$$

21.6% < 30%. Combining weak models drops performance significantly.

Voting Ensembles: The Cheat Sheet

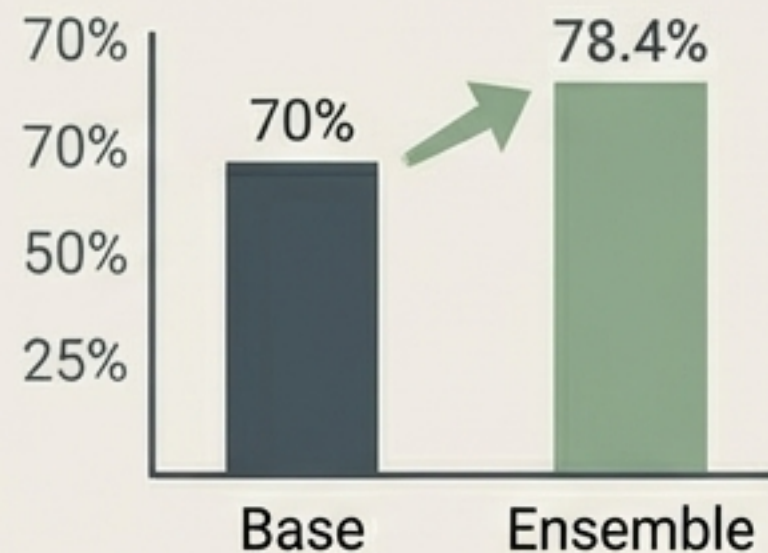
The Mechanics

Combines multiple base models. Uses Majority Voting for Classification and Mean for Regression.



The Math

Mathematically proven to increase accuracy by ensuring isolated individual errors are outvoted by a correct majority. (70% $\rightarrow$ 78.4%).



The Rules

Requires models to be completely Independent (diverse) and have a Base Accuracy strictly $>50\%$, otherwise it amplifies errors.

